Red alliance auto paths

BIOBUZZ 2026-27, Mechanical Mages. Values taken from the code on 2026-11-12.

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Match autos (Driver Station group LM1):
    RedAutoLM1                Shoot preloads, corner wall intake, far shot, top flower, far shot, park.
    RedLM12xFlowerAuto        Same as AutoLM1, but the side flower replaces the corner wall intake.
    RedTagAutoLM1             AutoLM1 plus a Limelight tag check after the preloads that picks route A or B.
    RedTag2xFlowerAutoLM1     LM12xFlower plus the same tag check after the preloads.
    RedFarWallAutoLM1         Wait at the far wall for our cell to tip, shoot 4, park.
    RedPreloadParkAutoLM1     Shoot the 4 preloads from the start, then park.

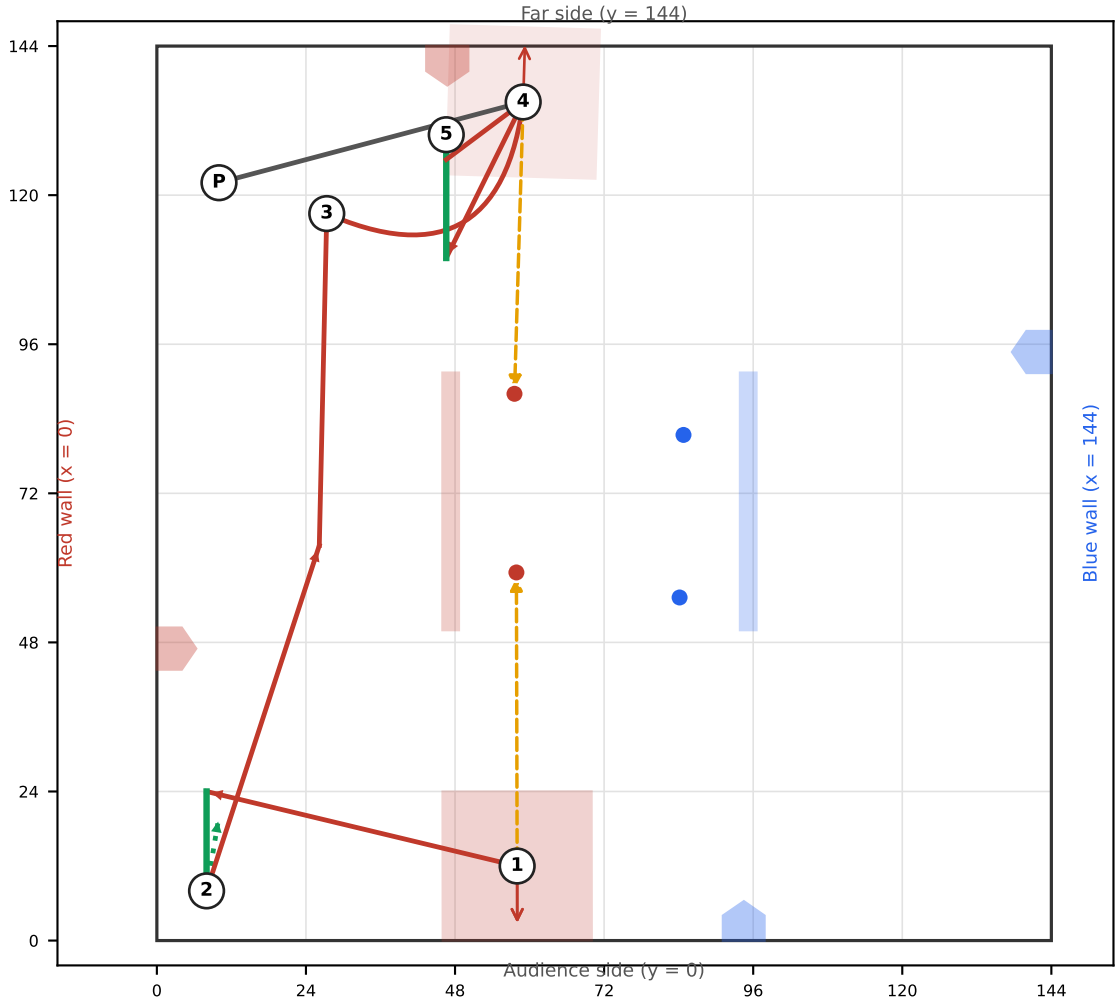
Field frame: inches, x from the red wall (0) to the blue wall (144), y from the audience side (0)
to the far side (144). Heading 0 = toward the blue wall, counterclockwise positive; heading is the
direction of the robot's FRONT (intake). The shooter is on the back.
Blue routes are the red routes rotated 180 degrees around the field center (72, 72).

Shared settings: shot speed 1530 ticks/s (FarWall uses the distance model - 50), fire feedforward
390 on the first shot and 290 after (boost only for the first 0.5 s), launcher voltage compensation on,
transfer turns slowly backward (-0.1) between shots after the first shot.

Other autos (no field map): SimpleMoveAutoLM1 (drive 12 in backward), SquareTestAutoLM1 (5 in strafe
square), PedroTest (Rishabh's single test path), and the old timed StarterBot autos (group StarterBot).
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RedAutoLM1

Shoot preloads, corner wall intake, far shot, top flower, far shot, park.



1. Start at (58.0, 12.0), heading -90, front against the audience wall. Turn in place to -90 deg (shooter at our audience-side cell) and shoot the 4 preloads at 1530 ticks/s (first-shot feedforward 390).

2. Corner wall intake: line up at (8.0, 24.0), push front-first into the corner at (8.0, 8.0), back out to (10.0, 20.0) and push again (2 pushes, 0.85 s each), intake on.

3. Up our wall, shooter first: (26.1, 63.6) (no stop) -> (27.4, 117.0), then curve (bend (55.0, 105.0)) to the far shot spot (59.0, 135.0).

4. Shot 2: turn so the shooter faces our far-side cell (88 deg), shoot (1 s min spin-up).

5. Top flower: drive to (46.6, 109.9), lower flower servo (1 s), drive in to (46.6, 129.7), slow push 2 in at 0.3, intake 1.75 s, back out 6 in to (46.6, 125.7), lift servo.

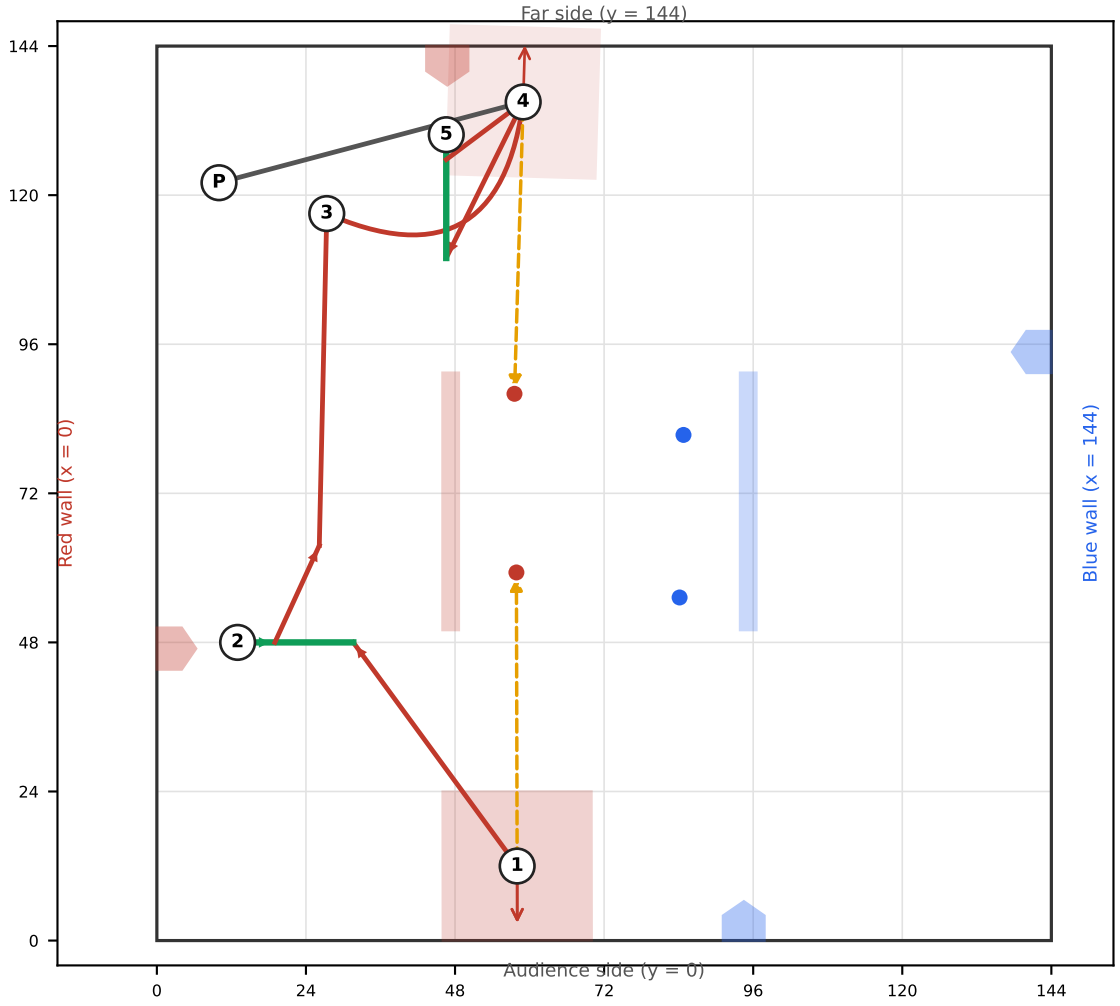
6. Back to (59.0, 135.0) and shoot again (shot 3).

P. Park: straight to (10.0, 122.0) (no turn).

Drive
Intake (thick) / back out (dotted)
Shot direction
Tag route B
Park

RedLM12xFlowerAuto

Same as AutoLM1, but the side flower replaces the corner wall intake.



1. Start at (58.0, 12.0), heading -90, front against the audience wall. Turn in place to -90 deg (shooter at our audience-side cell) and shoot the 4 preloads at 1530 ticks/s (first-shot feedforward 390).

2. Side flower: drive to (31.6, 48.0) facing the red wall, lower flower servo (1 s), drive in to (13.0, 48.0), slow push 2 in at 0.3 power, intake 1.75 s, back out 8 in to (19.0, 48.0), lift servo.

3. Up our wall, shooter first: (26.1, 63.6) (no stop) -> (27.4, 117.0), then curve (bend (55.0, 105.0)) to the far shot spot (59.0, 135.0).

4. Shot 2: turn so the shooter faces our far-side cell (88 deg), shoot (1 s min spin-up).

5. Top flower: drive to (46.6, 109.9), lower flower servo (1 s), drive in to (46.6, 129.7), slow push 2 in at 0.3, intake 1.75 s, back out 6 in to (46.6, 125.7), lift servo.

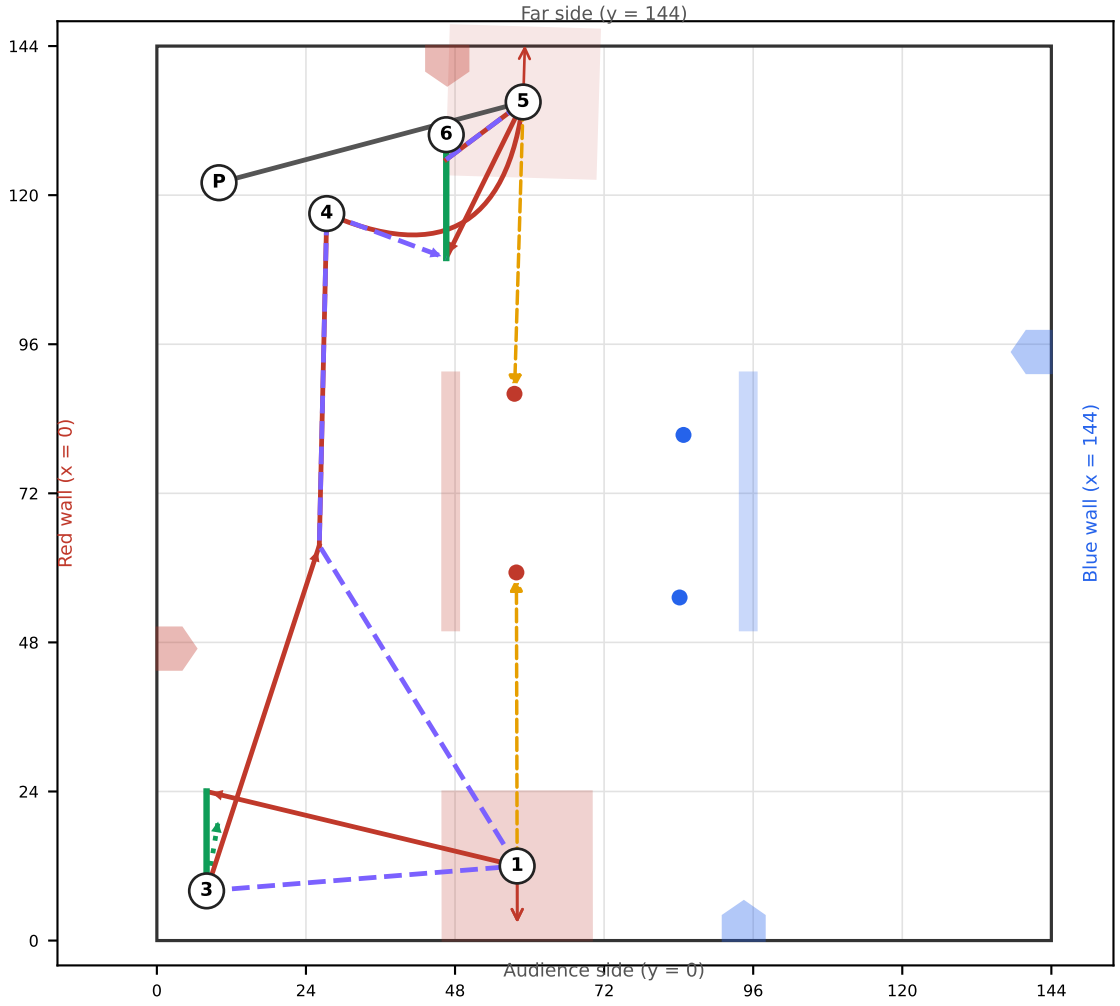
6. Back to (59.0, 135.0) and shoot again (shot 3).

P. Park: straight to (10.0, 122.0) (no turn).

Drive
Intake (thick) / back out (dotted)
Shot direction
Tag route B
Park

RedTagAutoLM1

AutoLM1 plus a Limelight tag check after the preloads that picks route A or B.



1. Start at (58.0, 12.0), heading -90, front against the audience wall. Turn in place to -90 deg (shooter at our audience-side cell) and shoot the 4 preloads at 1530 ticks/s (first-shot feedforward 390).

2. Tag check (3 s): stay aimed and watch our near cell's tags with the Limelight (tags 34-37). Tags GONE -> route A. Tags STILL THERE all 3 s -> route B.

Route A (solid):

3. Corner wall intake: line up at (8.0, 24.0), push front-first into the corner at (8.0, 8.0), back out to (10.0, 20.0) and push again (2 pushes, 0.85 s each), intake on.

4. Up our wall, shooter first: (26.1, 63.6) (no stop) -> (27.4, 117.0), then curve (bend (55.0, 105.0)) to the far shot spot (59.0, 135.0).

5. Shot 2: turn so the shooter faces our far-side cell (88 deg), shoot (1 s min spin-up).

6. Top flower: drive to (46.6, 109.9), lower flower servo (1 s), drive in to (46.6, 129.7), slow push 2 in at 0.3, intake 1.75 s, back out 6 in to (46.6, 125.7), lift servo.

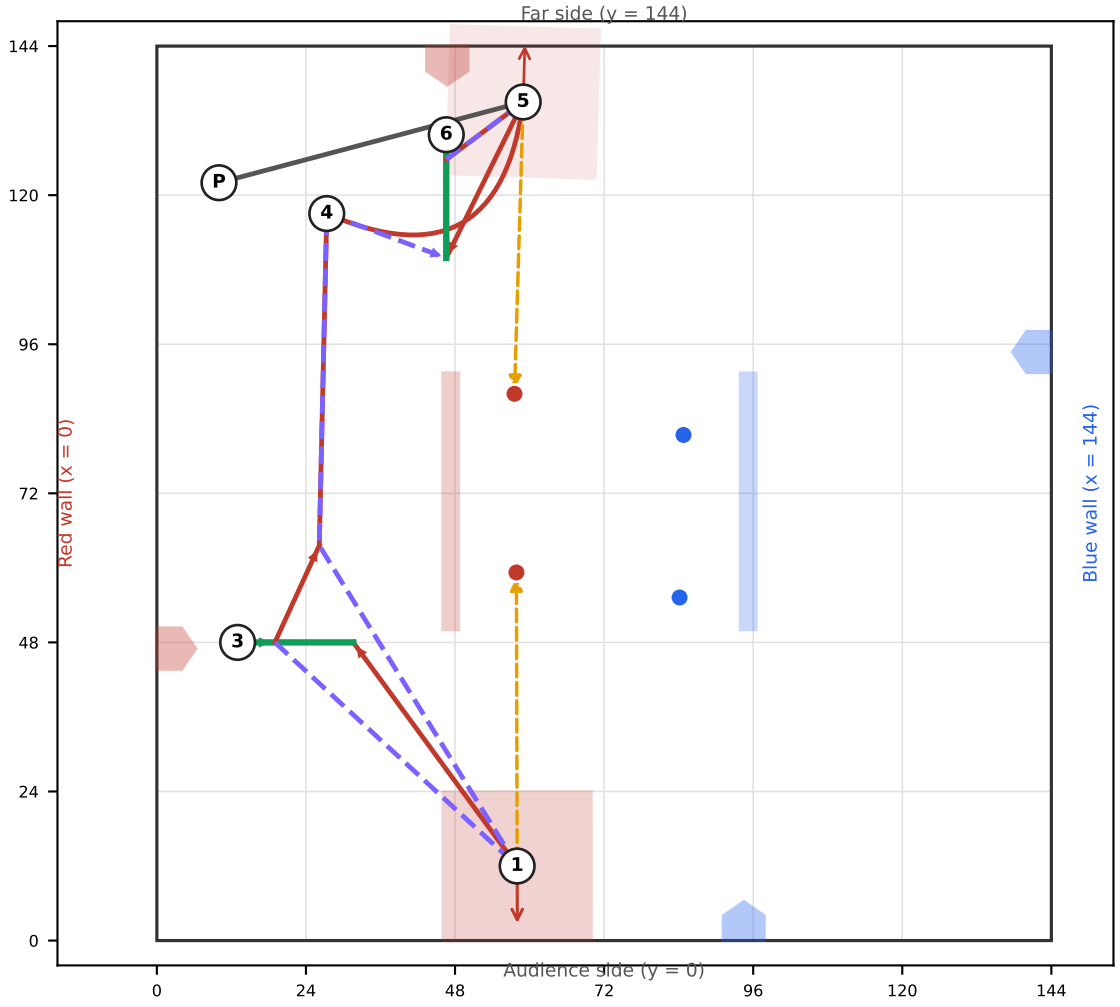
7. Back to (59.0, 135.0) and shoot again (shot 3).

P. Park: straight to (10.0, 122.0) (no turn).

Route B (purple dashed): same first intake -> back to start (58.0, 12.0) -> shoot the same cell again -> up our wall (26.1, 63.6), (27.4, 117.0) -> top flower (same steps) -> far shot spot (59.0, 135.0), shoot -> park (10.0, 122.0). Time checks: under 13 s left skips the 2nd start shot, under 18 s skips the top flower (it parks instead, up our wall).

RedTag2xFlowerAutoLM1

LM12xFlower plus the same tag check after the preloads.



1. Start at (58.0, 12.0), heading -90, front against the audience wall. Turn in place to -90 deg (shooter at our audience-side cell) and shoot the 4 preloads at 1530 ticks/s (first-shot feedforward 390).

2. Tag check (3 s): stay aimed and watch our near cell's tags with the Limelight (tags 34-37). Tags GONE -> route A. Tags STILL THERE all 3 s -> route B.

Route A (solid):

3. Side flower: drive to (31.6, 48.0) facing the red wall, lower flower servo (1 s), drive in to (13.0, 48.0), slow push 2 in at 0.3 power, intake 1.75 s, back out 8 in to (19.0, 48.0), lift servo.

4. Up our wall, shooter first: (26.1, 63.6) (no stop) -> (27.4, 117.0), then curve (bend (55.0, 105.0)) to the far shot spot (59.0, 135.0).

5. Shot 2: turn so the shooter faces our far-side cell (88 deg), shoot (1 s min spin-up).

6. Top flower: drive to (46.6, 109.9), lower flower servo (1 s), drive in to (46.6, 129.7), slow push 2 in at 0.3, intake 1.75 s, back out 6 in to (46.6, 125.7), lift servo.

7. Back to (59.0, 135.0) and shoot again (shot 3).

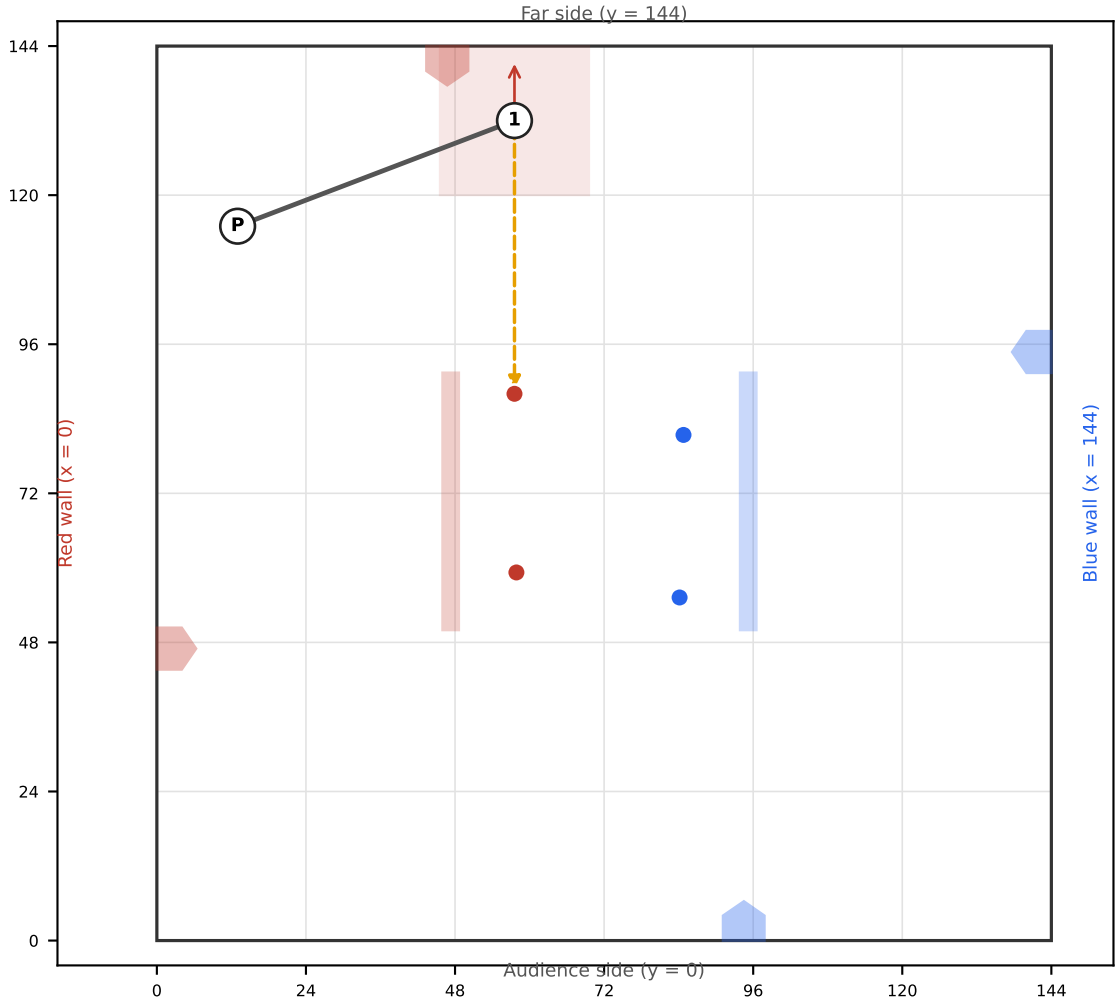
P. Park: straight to (10.0, 122.0) (no turn).

Route B (purple dashed): same first intake -> back to start (58.0, 12.0) -> shoot the same cell again -> up our wall (26.1, 63.6), (27.4, 117.0) -> top flower (same steps) -> far shot spot (59.0, 135.0), shoot -> park (10.0, 122.0). Time checks: under 13 s left skips the 2nd start shot, under 18 s skips the top flower (it parks instead, up our wall).

Drive | Intake (thick) / back out (dotted) | Shot direction | Tag route B | Park

RedFarWallAutoLM1

Wait at the far wall for our cell to tip, shoot 4, park.



1. Start with the front flat against the far wall at (57.6, 132.0) (heading 90), centered on our far-side cell, shooter facing the hive. Flywheel spins up (speed model - 50).

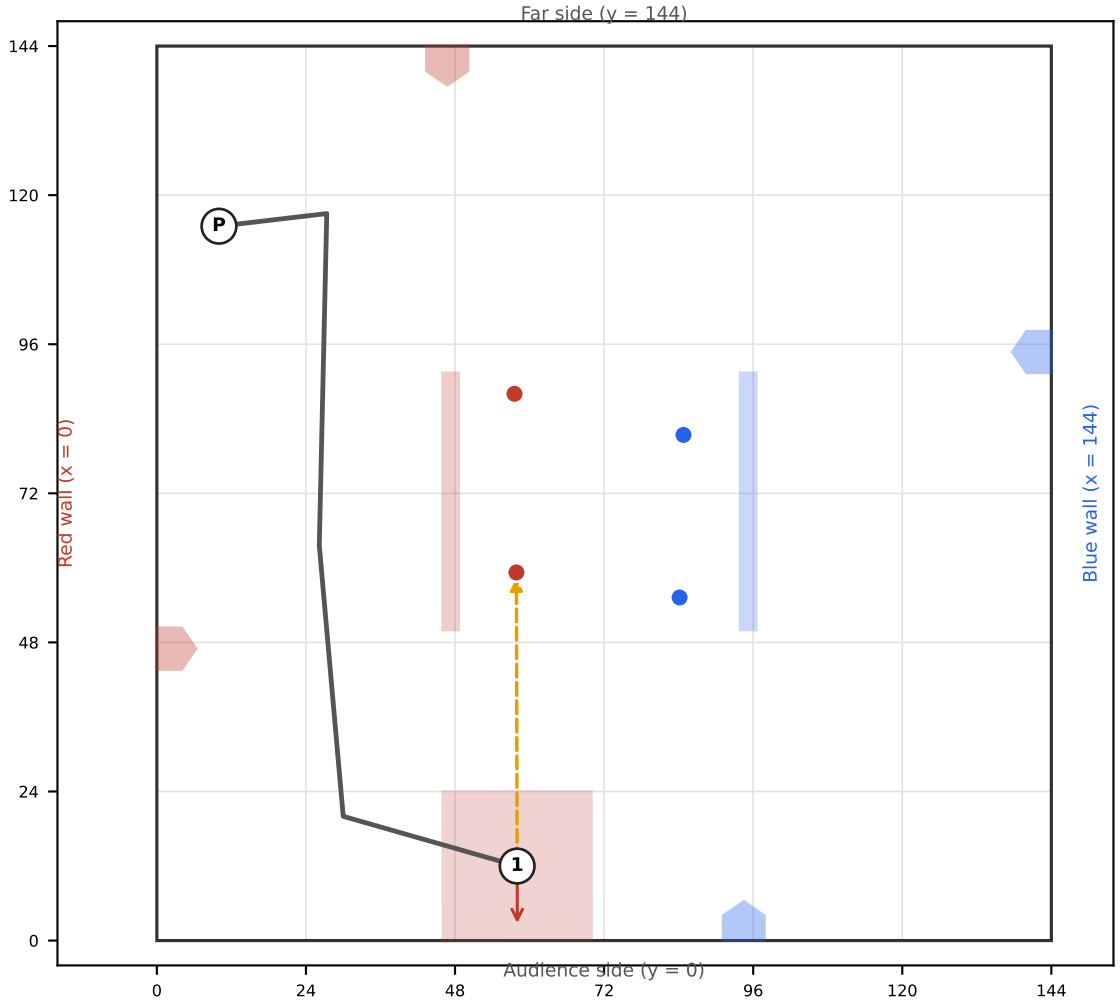
2. Wait for the cell to tip: as soon as the Limelight sees its tags (30-33), wait 3 s, then shoot all 4 without moving. No tags by 18 s: shoot anyway.

P. Park: drive to (13.0, 115.0).

Drive · Intake (thick) / back out (dotted) · Shot direction · Tag route B · Park

RedPreloadParkAutoLM1

Shoot the 4 preloads from the start, then park.



1. Start at (58.0, 12.0), heading -90, front against the audience wall. Turn in place to -90 deg (shooter at our audience-side cell) and shoot the 4 preloads at 1530 ticks/s (first-shot feedforward 390).

P. Park, shooter first (heading -90): (30.0, 20.0) -> (26.1, 63.6) -> (27.4, 117.0) -> (10.0, 115.0).

Drive
Intake (thick) / back out (dotted)
Shot direction
Tag route B
Park